

PRODUCT DESCRIPTION

FlexGrip is a one-component, clear, low-viscosity adhesive based on moisture-curing HydroCure Polymer technology powered by Covestro (Germany). The liquid cures by reacting with ambient humidity into a high-strength elastomer that combines rubber-like flexibility with rope-like resistance to tearing: the cured film stretches to 460 % of its original length while tensile strength climbs to 33 MPa at break. Because the bond flexes with the substrate instead of resisting it, FlexGrip survives the repeated flexing that cracks rigid adhesives — the property that allows it to replace stitching in shoe-sole repair, its original application. It bonds without abrasion or primer to rubber, leather, ceramic including glazed tile, wood, metal, brick and most plastics, withstands permanent water immersion after full cure, and is deliberately slow-setting so that parts can be repositioned without time pressure during assembly.

KEY DATA

Chemistry	One-component moisture-curing HydroCure Polymer — powered by Covestro (Germany)
Supplied form	Clear, water-thin liquid — ready to use; no mixing, no catalyst, no primer
Cure mechanism	Reacts with humidity in the air and substrate; deliberately slow-setting to allow repositioning during assembly
Tensile strength	33 MPa at break (cured free film, JIS K 6251) — approx. 330 kg per cm ² of bond area
Elongation at break	460 % (cured free film, JIS K 6251)
Tensile modulus	8.7 MPa at 100 % elongation · 18 MPa at 300 % elongation
Adhesion	Rubber, leather, ceramic incl. glazed tile, wood, metal, brick, most plastics — no abrasion, no primer required
Application	Thin, full-coverage film on one or both faces; parts held in firm, continuous contact throughout the clamping period
Surface set	Approx. 6 hours
Clamping time	12 hours minimum before releasing clamps, bands or tape
Full cure	Approx. 40 hours before heavy service, hard use or water contact
Water resistance	Permanent immersion after full cure; bonding to damp and underwater surfaces is possible where unavoidable — a clean, dry surface gives the highest bond strength
Chemical resistance	Cured film resists water, dilute acids and common solvents including thinner
UV behaviour	Bond strength is unaffected by sunlight; exposed cured adhesive may yellow over time — keep bond lines thin and closed with no squeeze-out (see notes)
Colour	Clear
Packaging	200 g bottle · 20 ml trial sachet
Storage	Sealed original container, dry and shaded, room temperature — do not refrigerate
Shelf life	12 months unopened — see storage notes; the opened bottle is moisture-sensitive

NOTE ON VALUES

Mechanical values are measured on a fully cured free film to JIS K 6251 and describe the adhesive material itself. The strength of a finished joint additionally depends on surface cleanliness, film thickness, contact pressure during cure and the bonded area — a thin, fully wetted film held in firm contact for the full clamping period reaches the highest joint strength. Figures are typical guidance only and are not a specification.

DESIGN PRINCIPLE — FLEXIBLE, NOT HARD

A shoe sole flexes thousands of times a day. A rigid, glass-hard adhesive resists that movement, concentrates stress at the bond edge and fails by cracking within weeks — the standard failure mode of hot-melt and contact adhesives in sole repair. FlexGrip takes the opposite route: the cured film is a high-grade elastomer whose tensile strength rises as it stretches, from 8.7 MPa at 100 % strain to 33 MPa at the 460 % break point. The bond therefore follows every flex of the substrate while the reserve of strength grows under load — which is why a glued repair can genuinely replace stitching, and why the same behaviour carries over to loose tiles, cracked fan blades, brick, battery casings and the many other repairs users have taken it to.

APPLICATION NOTES

- Clean the surfaces and let them dry: remove dust, soil and grease completely. No abrasion and no primer are required, but do not apply to a wet surface unless unavoidable — moisture on the bond face reduces performance for no benefit.
- Apply a THIN film over the full contact area, on one or both faces. Never apply a thick, oozing bead. If excess squeezes out of the joint, wipe it off immediately while still wet — once cured it cannot be removed by any solvent.
- Wide gaps and large voids (e.g. a debonded tile): lay a piece of tissue paper soaked in adhesive into the gap. This increases the bonded area, saves adhesive and produces a tighter repair.
- Clamp the parts in firm, continuous contact — rubber bands, tape, clamps or a dead weight, whatever is at hand. Parts can still be repositioned without hurry during the open time.
- Shoe repairs: always pack the shoe to shape with cloth or paper BEFORE binding or taping. Binding an empty shoe distorts it permanently — no adhesive can correct a shoe cured out of shape.
- Respect the clock: surface set approx. 6 hours; keep the joint clamped for a full 12 hours before releasing — releasing early is the leading cause of failed repairs; allow approx. 40 hours before heavy service or water contact.
- Adhesive on skin: scrub the hands with DRY detergent powder, without water, until the tack is gone, then rinse; repeated washing-up liquid is the fallback. Never clean skin with thinner or strong acid — hazardous to the skin, and the cured adhesive resists thinner in any case.
- Bottle care — humidity is the enemy: after use, wipe the bottle neck and cap clean (otherwise the cap seizes), tape over the neck as an extra air seal, then close the cap tightly. Store dry, shaded, at normal room temperature. Do not refrigerate.

HEALTH & SAFETY

Uncured adhesive contains an aromatic isocyanate prepolymer and may cause skin and respiratory sensitisation. Work in a ventilated area; wear protective gloves and eye protection; avoid skin and eye contact with uncured material. In case of eye contact, rinse immediately with plenty of clean water and seek medical attention. Fully cured material is inert. Keep out of reach of children. Refer to the Safety Data Sheet for full information.

STORAGE

Store in the original tightly sealed container in a dry, shaded place at normal room temperature. The product cures on contact with humid air — air entering an opened bottle is what spoils the adhesive before its time. Reseal immediately after use as described in the application notes. Do not refrigerate. Shelf life 12 months unopened.